AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 22 · STUDENT LESSON

Missing and Unbalanced Examples

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

Count distinct cases. Inspect the missing conditions.

Name target collections, inspect selection frames, compare coverage plans and audit a new sound-recognition claim.

Keep one habit: State the target → inspect crossed conditions → separate selection from assessed outcomes.

MAIN LESSON · SECTION 1

Which intended cases have actually been checked?

> Name the target, selection list, distinct cases and assessed conditions.

Start with more files but the same cards

Twelve image files can still represent only six distinct cards. Count the underlying cases and the conditions they cover before using a large-looking total as evidence.

Retrieve schema validity versus coverage

A table has been checked and every recorded value matches its source. Does that prove it includes all the conditions in which a system will be used? Explain.

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

By the end, you can...

- I can name the target collection, selection list and cases actually assessed.
- I can use a **coverage** table to identify missing conditions before making a claim.
- I can propose a selection plan that addresses a stated gap without promising certainty.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Good Data, Bad Data. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Similarity. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Boundaries and 'Not Sure'. Give an example and a mistake that the idea helps you avoid.

What is absent can shape a system's failures.



LEARNING TARGET A **training** set has 18 daylight photos and two dark photos of one object. Adding eight extra copies of each dark photo makes 18 dark-image files, matching the 18 daylight files. Does it balance the variety? Design a better collection step.

Learner name:

Readiness for the connected explanation

This week’s end goal

By the end, I can identify missing conditions and design a targeted **coverage** check.

MAIN LESSON · SECTION 2

Separate the target collection from its sample and frame

> Shapes24 contains all four shape-by-lighting combinations, each with six distinct IDs.

Read the full target and checked labels

Suppose a shape recogniser will be evaluated on a fixed collection of 24 invented picture cards. This **target population** has 6 bright circles, 6 dim circles, 6 bright squares and 6 dim squares.

Each card has its own ID and an independently checked circle or square label. We want to know how often the recogniser gives the correct label across these 24 cards.

State the original sample and its limited result

A tester takes the 6 bright-circle cards nearest the front of the box. Those 6 cards form the **sample**; the other 18 have not been assessed. The recogniser is correct on 5 of the 6 **sample**d cards.

That is a result for these assessed bright circles. It does not establish correctness on the squares or dim cards.

Inspect the list used for selection

A **sampling frame** is the list or source available for selection. A complete list of all 24 IDs could give every card a chance of selection.

A list containing only bright-circle IDs cannot select dim cards or squares, even if the choice from that list is random. These fictional picture cards do not contain personal information.

Explain the coverage gap

- Name the 24-card target and the original six-card sample. Which cases can a bright-circle-only list never select? Explain why random choice within that incomplete list does not add the missing conditions.

▼ Earlier wording, alternatives and picture transcripts

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



MYSTERY An animal recognizer trained only on bright daytime photos may fail on the same animals at night.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first **prediction** and state the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

MAIN LESSON · SECTION 3

Check combinations as well as individual labels

> Having both shapes and both lighting labels somewhere can still leave crossed conditions absent.

Read the condition-by-condition coverage

Coverage is about cases as well as counts

CARD CONDITION	TARGET COUNT	ASSESSED COUNT
Bright circles	6 cards	6 assessed
Dim circles	6 cards	0 assessed
Bright squares	6 cards	0 assessed
Dim squares	6 cards	0 assessed

All 6 assessed cases come from one of four conditions. An assessed count of zero means no evidence for that condition; it does not mean the recogniser failed every case in it.

Distinguish image files from underlying card IDs

The target has 24 distinct cards, but the **sample** covers only the 6 bright circles. Photographing each **sample**d card twice would create 12 image records without adding a new card or covering a missing condition.

Record the identity of the underlying case as well as the number of files.

Inspect the missing combinations

Both shape and lighting matter to this question. A **sample** with bright circles and dim squares includes each shape and each lighting label, yet still misses dim circles and bright squares.

Check combinations when the interaction could matter, not just whether each individual label appears somewhere.

Explain a table entry

- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

VISUAL **MECHANISM**

Input → Process → Output

Follow the information instead of imagining magic inside the machine.



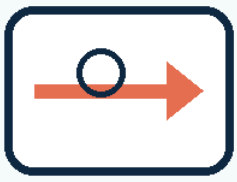
INPUT

example set with represented and absent conditions



PROCESS

learn patterns from available coverage



OUTPUT

uneven performance across conditions

INPUT example set with represented and absent conditions

PROCESS learn patterns from available **coverage**

OUTPUT uneven performance across conditions

CORE IDEA What is absent can shape a system's failures.

UNDER THE HOOD Sampling **imbalance** changes the empirical distribution seen during **training**. Missing **subgroup**s or conditions cannot contribute error signals, so average success may hide systematic failure.

Rebalancing requires relevant examples, not arbitrary duplication alone.

Draw a fresh example with all three stages:

A closer explanation

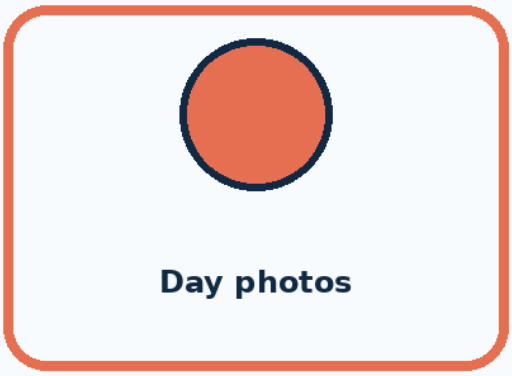
Connect the picture and the explanation

MAIN LESSON · SECTION 4

Compare coverage and variety clues

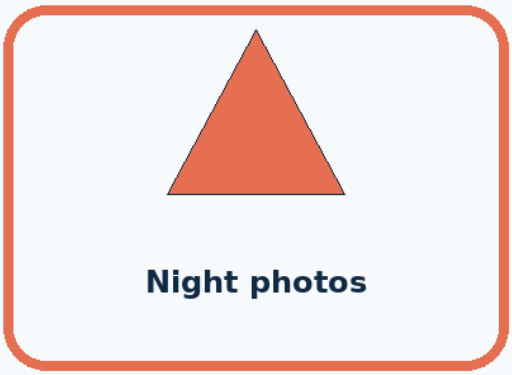
> Ask which new conditions and distinct cases an added record actually contributes.

Pictures paired with their evidence



The case to investigate

CASE 1 — Day photos Clue: many clear examples. Meaning: strong represented condition.



The case to investigate

CASE 2 — Night photos Clue: none available. Meaning: **coverage gap**.



The case to investigate

CASE 3 — Cats 90, dogs 10 Clue: uneven class counts. Meaning: class **imbalance**.



The case to investigate

CASE 4 — One dog copied 20 times Clue: count rises, variety does not. Meaning: weak repair.

Discuss the confusing pair

- Which two pictures are easiest to confuse? What **evidence** separates them?

Keep the limit in view

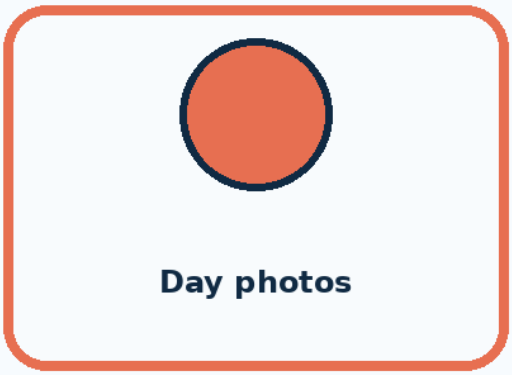
BOUNDARY Balance is about useful representation, not forcing every count to be identical without reason.

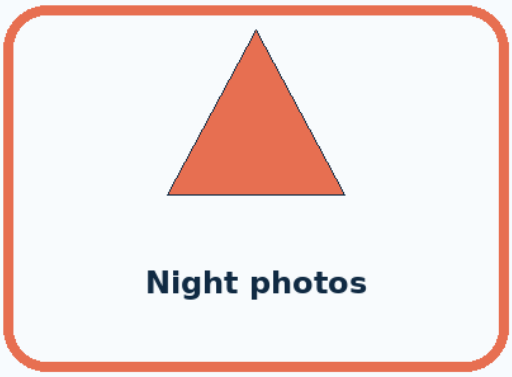
▼ **Earlier wording, alternatives and picture transcripts**

PICTURE GALLERY

Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.









CASE 1 — Day photos Clue: many clear examples. Meaning: strong represented condition.

CASE 2 — Night photos Clue: none available. Meaning: **coverage gap**.

CASE 3 — Cats 90, dogs 10 Clue: uneven class counts. Meaning: class **imbalance**.

CASE 4 — One dog copied 20 times Clue: count rises, variety does not. Meaning: weak repair.

Change one thing in the pictured case

Change: Add the two remaining bright plain cards and no dim cards.

Prediction: The **sample** becomes larger but still has no dim **coverage**.

Reason: The missing lighting condition remains untested.

MAIN LESSON · SECTION 6

A larger bright-only sample still misses dim conditions

> This separate Badges32 collection has eight target cards in each lighting-by-pattern group.

The target and development sample

A fictional badge reader will be used in both bright and dim rooms.

Intended target has 8 bright plain,8 bright patterned,8 dim plain,8 dim patterned cards.

A development **sample** contains 6 bright plain cards and 2 bright patterned cards; no dim cards.

Read the four group counts



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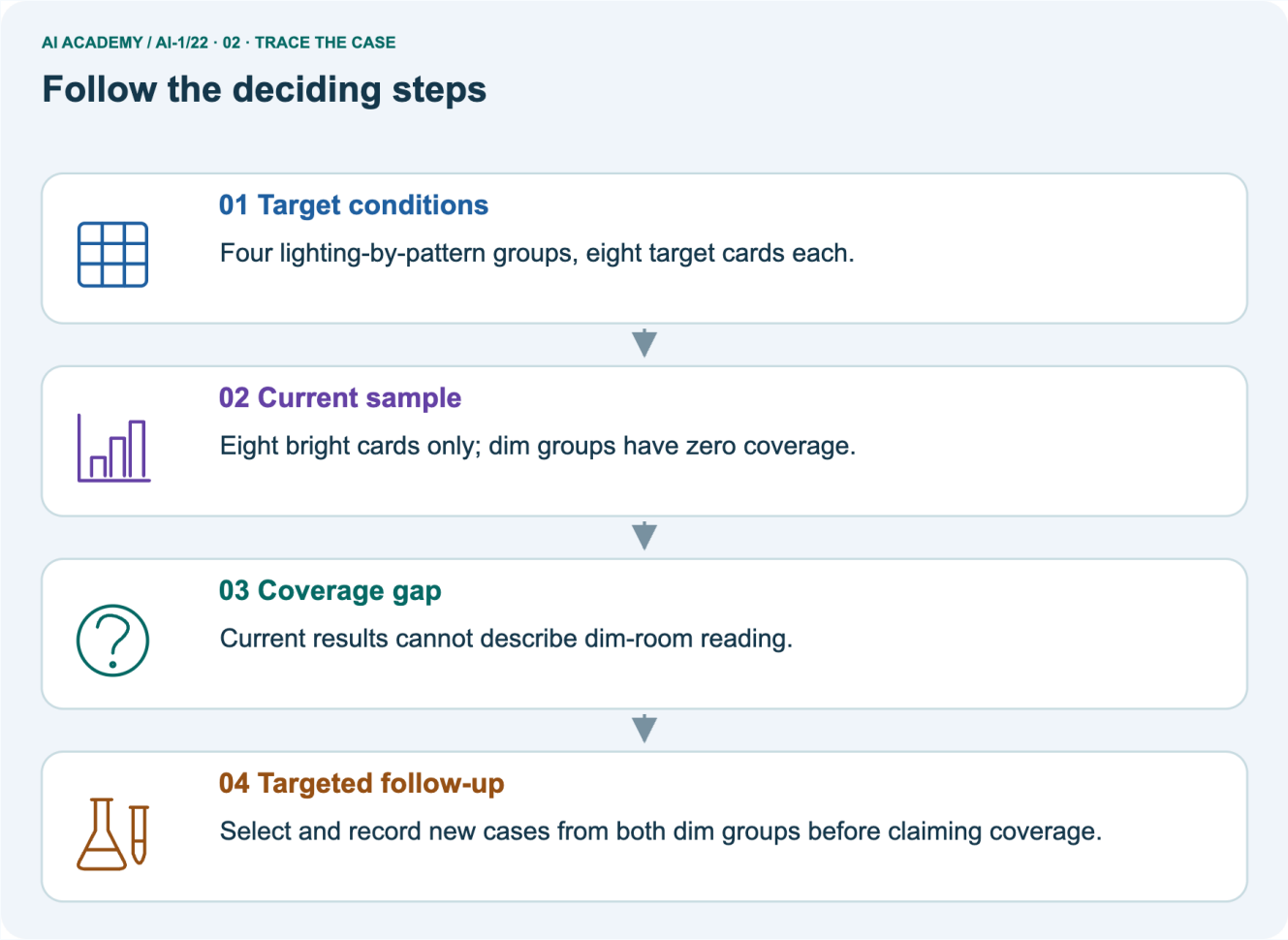
Target groups and development coverage

	Target cards	Sample cards
Bright plain	8	6
Bright patterned	8	2
Dim plain	8	0
Dim patterned	8	0

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

Trace the targeted coverage follow-up



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain why size and coverage differ

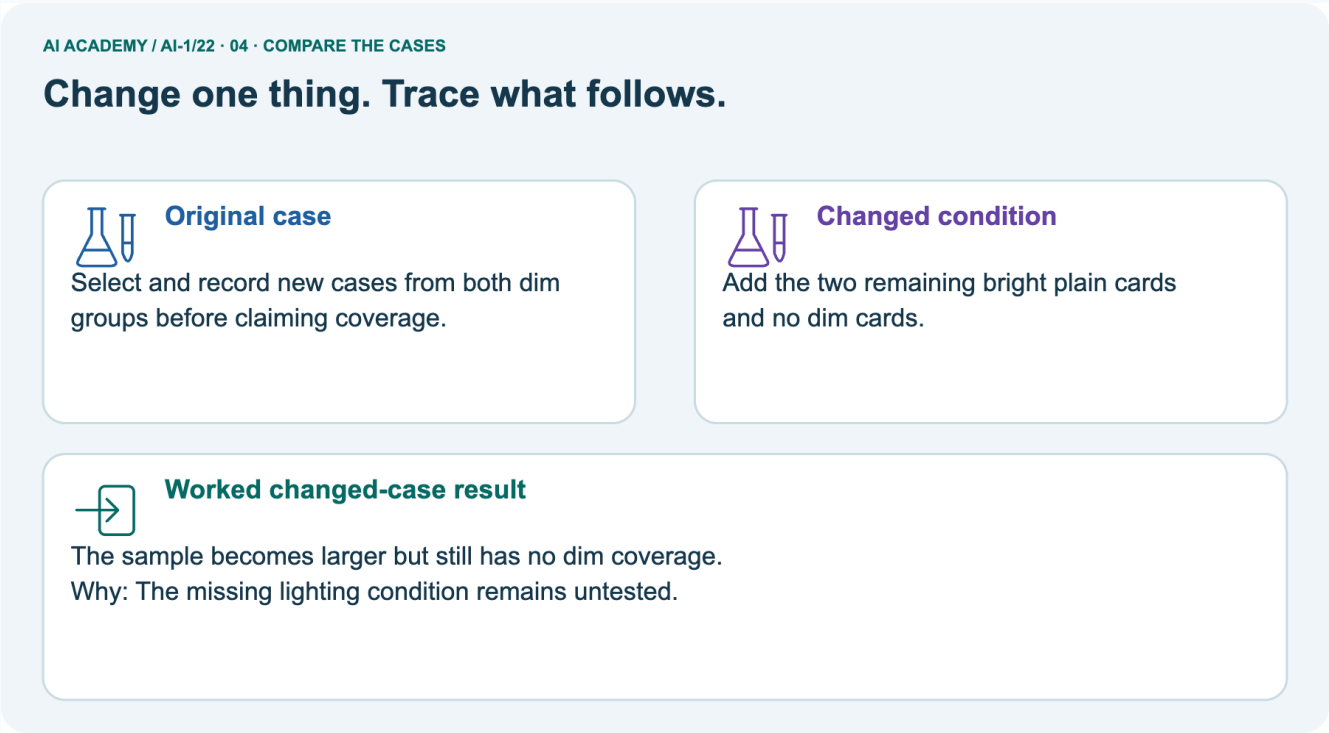
Adding more bright cards can increase **sample** size without addressing the missing condition. A **coverage** check asks which relevant situations are absent, not merely how many rows exist.

Deliberately selecting dim cases helps investigate those conditions, but the resulting test should not be advertised as a random survey unless its selection really was random.

Predict the larger bright-only sample

- Before checking, add the two remaining bright-plain cards to the badge sample and no dim cards. Recount sample size and identify the condition groups that still have zero coverage.

▼ Check the remaining gap



Follow the labelled picture with the matching explanation. Open the vector diagram

Limit what targeted coverage establishes

Coverage does not guarantee accuracy, and targeted cases alone do not supply a population-wide success estimate.

Inspect whether one dim group is enough

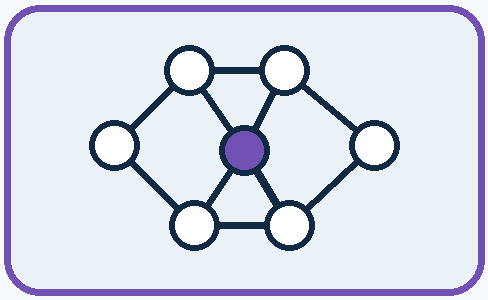
- Read the labelled picture: Would testing only dim plain cards close every current gap? Point to the target groups.

▼ Earlier wording, alternatives and picture transcripts

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



CASE An animal recognizer trained only on bright daytime photos may fail on the same animals at night.

STEP 1 — OBSERVE example set with represented and absent conditions

STEP 2 — TRACE learn patterns from available **coverage**

STEP 3 — CONCLUDE uneven performance across conditions

MODEL REASONING The model may rely on bright backgrounds or fail to detect shapes in low light. Collect varied night examples for every relevant animal and preserve a separate night test set to evaluate the repair.

DO NOT OVERCLAIM Balance is about useful representation, not forcing every count to be identical without reason.

What new **evidence** could change this answer?

A closer explanation

Plan	What it helps with	Limit to report
Take cards at the front	Quick initial check	Front position may restrict conditions
Draw from all 24 IDs by chance	Every listed card can be selected	A small draw need not balance conditions
Choose 2 distinct cards per condition	Includes all four conditions	8 cards do not establish all 24 outcomes
Assess all 24 distinct cards	Covers this entire fixed collection	Does not prove performance on future images

For a broad estimate, a chance-based plan should start from a suitable complete list and use a documented selection procedure.

For investigating missing conditions, deliberately selecting cases from each condition can be useful. State which purpose you chose. A set built to expose problems is not automatically a faithful miniature of everyday use.

Selection bias is about a systematic feature of the selection **process**, not whether a person had bad intentions. Choosing only easy-to-reach, bright cards can leave harder conditions out.

More cases from the same narrow source may repeat that gap. A larger **sample** cannot by itself repair a frame that excludes relevant cases.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Missing and Unbalanced Examples

The facts we will use

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Target groups and development coverage

	Target cards	Sample cards
Bright plain	8	6
Bright patterned	8	2
Dim plain	8	0
Dim patterned	8	0

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

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Missing and Unbalanced Examples

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Target conditions

Four lighting-by-pattern groups, eight target cards each.



2. Current sample

Eight bright cards only; dim groups have zero coverage.



3. Coverage gap

Current results cannot describe dim-room reading.



4. Targeted follow-up

Select and record new cases from both dim groups before claiming coverage.

CHANGE ONE THING

Add the two remaining bright plain cards and no dim cards.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Target conditions	Four lighting-by-pattern groups, eight target cards each.
2. Current sample	Eight bright cards only; dim groups have zero coverage.
3. Coverage gap	Current results cannot describe dim-room reading.
4. Targeted follow-up	Select and record new cases from both dim groups before claiming coverage.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

Change one thing: Add the two remaining bright plain cards and no dim cards. Predict the result before checking the explanation. What remains the same?

Report the target and the reach of the evidence

> State the assessed cases, absent conditions and selection method.

Keep the original five-of-six result bounded

A careful report identifies the target, frame, selection method, distinct cases, assessed outcomes and gaps.

For the original **sample**: 5 of 6 selected bright-circle cards were correct; the 18 cards in other conditions were not assessed. This is useful **evidence** with a narrow reach, not a score for all future images.

Revisit coverage when the intended use changes

Even checking all 24 cards answers a question about that fixed collection and that recogniser version. New image sources, different lighting or other shapes introduce new cases.

Revisit the target and the **coverage** plan when the intended use changes.

Repair the 100-more-bright-images claim

- CLAIM A A model can infer missing conditions automatically.
- CLAIM B Copying rare examples many times creates new variety.

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

> **A tester collects 100 more bright-circle images and says the absence of dim-square tests no longer matters. Explain what increased and what coverage gap remains**

▼ Earlier wording, alternatives and picture transcripts

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case	Evidence	What it shows
Day photos	many clear examples	strong represented condition
Night photos	none available	coverage gap
Cats 90, dogs 10	uneven class counts	class imbalance
One dog copied 20 times	count rises, variety does not	weak repair

Repair two tempting claims

Repair:

Repair:

A closer explanation

MAIN LESSON · SECTION 8

Practise: selected is different from assessed

> An unavailable file is recorded separately when the stated task measures recognition correctness only.

Inspect the eight-selected, seven-assessed example

Imagine a new plan selects 2 distinct cards from each of the four conditions: 8 cards in total. One selected dim-square file cannot be opened. Only 7 have assessed outcomes.

Record the missing file and its condition; do not call it a recognition error or pretend all 8 outcomes were checked. This exercise evaluates recognition only, so file availability is reported separately.

Rewrite the original overly broad claim

> **A report says the original recogniser is correct on 5 of 6 cards, so it works on all shapes and lighting. Rewrite the claim using the original sample, identify the three missing conditions, and propose a selection plan to investigate them**

Keep coverage separate from test independence

Choose and document cases before seeing their outcomes. If you use the results to revise the recogniser, keep a separate unseen final test as in Week 15.

Coverage and an unseen test solve different problems: one asks which conditions appear; the other asks whether the evaluation cases influenced the revision.

Use W2 to count distinct cards

Use workbook W2 for the connected example “Sampling and **coverage**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Challenge “more records always remove bias”

- Use the worked case for Sampling and **coverage**. Proposed explanation: “More records always remove bias”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Return to the two dim badge groups

Would testing only dim plain cards close every current gap? Point to the target groups.

> **My explanation using a pictured step and the given facts**

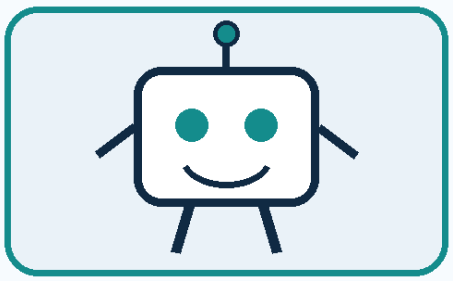
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: What is absent can shape a system's failures.

Use a fresh example to explain **missing example**.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: Copying rare examples many times creates new variety.

Invent one edge case and name the safe response.

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

Practise with less help: Separate files from distinct cases

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Investigate missing animal-photo conditions

> Use a complete fictional target list and distinguish a coverage plan from observed model performance.

The target IDs and development coverage

Use a fictional 12-card animal target with supplied reference labels. Conditions and distinct IDs are day fox FD1–FD3, night fox FN1–FN3, day owl OD1–OD3, night owl ON1–ON3. Development used the six day cards only. Recognition outcomes for the six night cards have not been obtained. Repeated copies keep the same underlying card ID.

Compare copies with new condition coverage

1. Predict which conditions need investigation without declaring their recognition outcomes in advance. Make a coverage table. Compare adding six copies of day cards with reserving two distinct night-fox and two distinct night-owl IDs for a targeted check of the frozen recognizer. State exactly what the selection plan checks, which night IDs remain unused, and which outcomes are still unknown. Label this paper work as a selection audit, not a completed recognition experiment.

2. Keep a normal case and an edge case clearly separated.

3. Record what happened—even when it does not match your **prediction**.

4. Explain the result, one limitation, and the next useful test.

Keep selection, assessment and unknown outcomes separate

Case	Prediction	Observed result	What it means
Normal			
Edge			

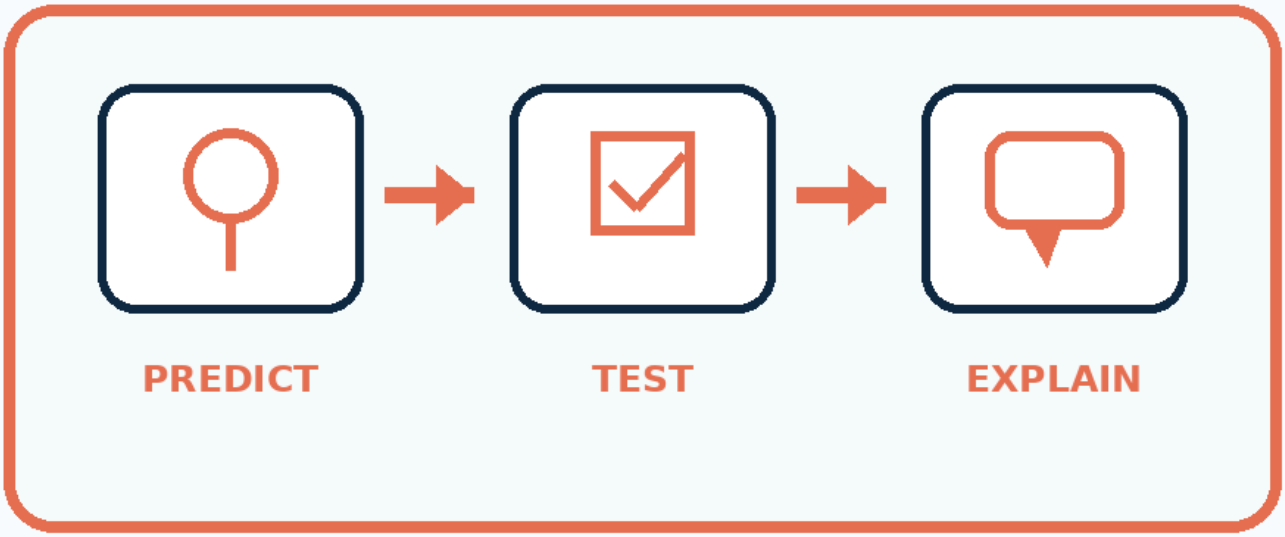
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

EXPLORER LAB

Missing-Condition Detective

Predict first, test honestly, and explain what the **evidence** means.



State your **prediction** before the result is known.

Complete the task: Predict failures in an animal set missing night photos.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: audit the new sound-recognition claim

> Audit the original result and propose a labelled coverage investigation; record any reused clip IDs.

The new target, sample and requested plan

A fixed target has 20 invented clips: 5 quiet bells, 5 noisy bells, 5 quiet claps and 5 noisy claps. A tester selects 4 distinct quiet bells from a quiet-bell-only list; 3 are correctly labelled and 1 is wrong.

Write a supported result, name missing conditions and explain why the result is not established for all 20 clips. Propose an 8-clip plan that includes every condition, with no clip selected twice.

> **My independent explanation, trace and limits**
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>
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Keep one first attempt

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Audit a new sound-recognition claim

Week 22 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A photo sorter trains on ten bright photos and no dim photos. What group of examples is missing? Design a test that reveals the gap.

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

List normal, low-light, partly hidden, worn, and unusual items; mark which conditions are missing from the sorter data.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Transfer practice from the original programme

A **training** set has 18 daylight photos and two dark photos of one object. Adding eight extra copies of each dark photo makes 18 dark-image files, matching the 18 daylight files. Does it balance the variety? Design a better collection step.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Build a crossed-condition coverage table for your sorter

> List important conditions, distinct IDs and the groups actually assessed.

This week’s project decision

THIS WEEK'S CHECKPOINT List normal, low-light, partly hidden, worn, and unusual items; mark which conditions are missing from the sorter data.

Create a sorter **coverage** table crossing important classroom conditions, then state which groups each test actually represents.

Plan conditions, selections and checks

- What real classroom need does this serve?
- What information is necessary—and what should stay private?
- What happens inside the **process**?
- What normal and edge cases will you test?

- When should it say not sure or ask a person?

Choose representation for the stated purpose

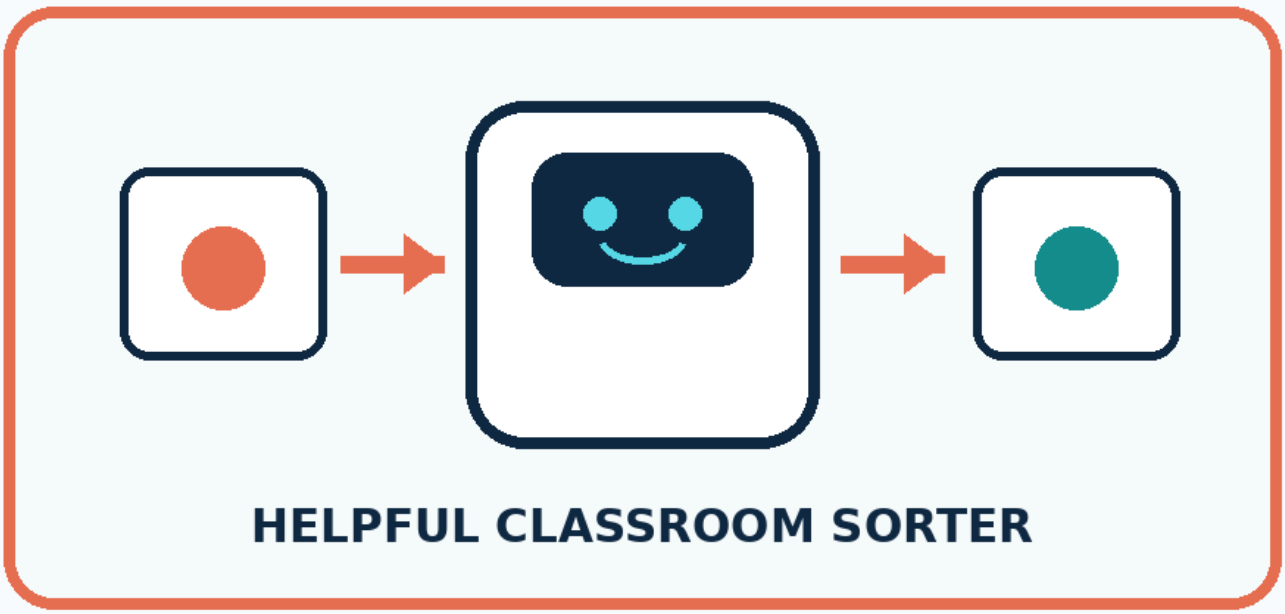
RESPONSIBLE DESIGN Balance is about useful representation, not forcing every count to be identical without reason.

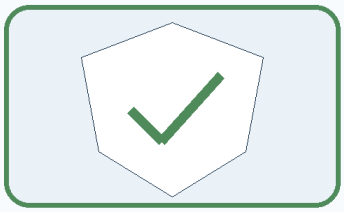
▼ **Earlier wording, alternatives and picture transcripts**

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.





My project decision:

Evidence for your project

MAIN LESSON · SECTION 16

Finish: name what the evidence does and does not cover

> Close your notes and identify the target, frame, distinct selected cases and assessed outcomes.

Your exit explanation

- Close your notes. Explain this goal in your own words: Name the target collection, selection list and cases actually assessed

> **My deciding step, evidence and one remaining question**

Choose support and revisit later

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	Meaning	My example
missing example	needed condition absent from data	
imbalance	some groups appear much more than others	
subgroup	defined part of the intended population	
coverage gap	important situation poorly represented	

Terms for the connected explanation

Word	Meaning
Target population	The whole collection of cases a question is about
Sample	The selected cases used to investigate a larger collection
Sampling frame	The list or source from which cases can be selected

Word	Meaning
Coverage	Which relevant kinds of cases the list or sample includes
Selection bias	Systematic distortion caused by how cases enter an investigation

Repair a claim using a word

- CLAIM TO REPAIR A model can infer missing conditions automatically.
- Use one vocabulary word and one piece of **evidence** to repair it:

▼ Earlier wording, alternatives and picture transcripts

VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.









Misconception detective

WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

▼ Choose a stronger thinking challenge

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

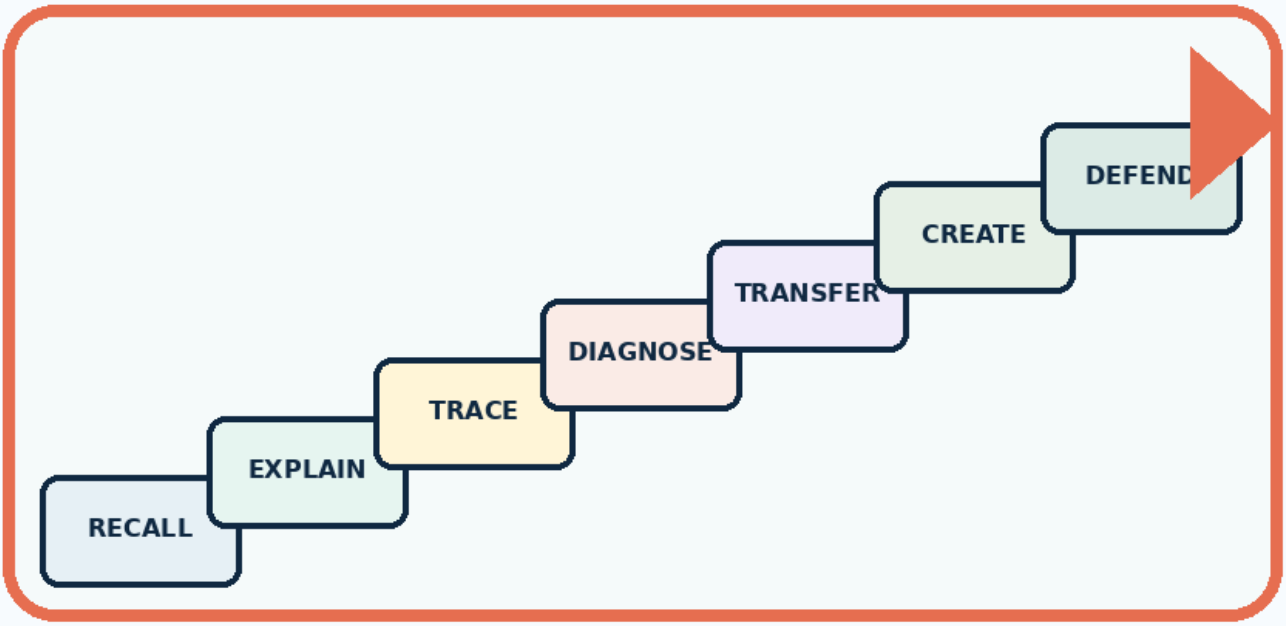
- RECALL Define **missing example**.
- EXPLAIN What is absent can shape a system's failures.
- TRACE Show the **input**, **process**, and **output**.
- DIAGNOSE Repair: A model can infer missing conditions automatically.
- TRANSFER Apply the idea to a new home or classroom case.
- CREATE Design one revealing test.
- DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

▼ Earlier wording, alternatives and picture transcripts

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

▼ **Reference: check your explanation**

REFERENCE · SECTION 12

Reference: check your explanation

> Use after practice. Audit a new sound-recognition claim is the selected independent case.

Check the evidence and its limit

- ☐ Name or draw the fresh case.
- ☐ Trace **input**, **process**, and **output**.

- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Record support

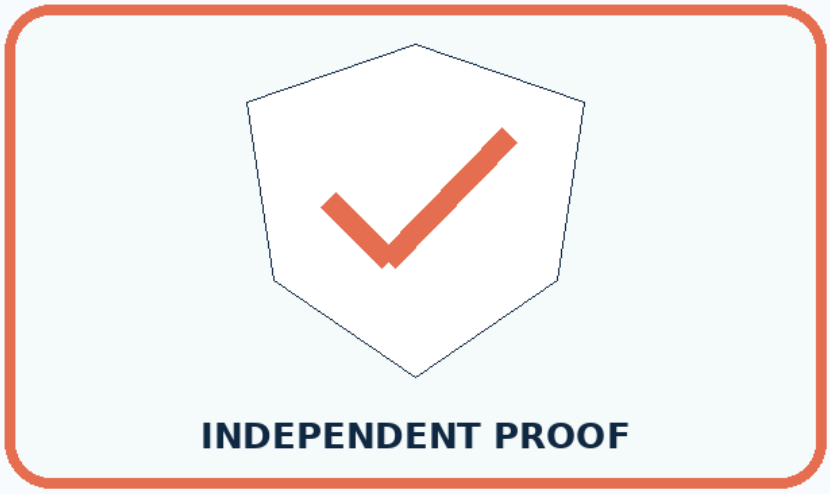
My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

▼ Earlier wording, alternatives and picture transcripts

MASTERY

Checklist for explaining this week's idea

Use after practice. Audit a new sound-recognition claim is the selected independent case.



EXPLANATION CHECK Name the target and frame, count distinct cases and crossed conditions, distinguish selected from assessed outcomes and explain the purpose and limits of a follow-up plan.

Independent **evidence**:

SELF-CHECK The model may rely on bright backgrounds or fail to detect shapes in low light. Collect varied night examples for every relevant animal and preserve a separate night test set to evaluate the repair.

NEXT BRIDGE Carry today's **evidence** habit into: Fair for Whom?.

▼ **Extra practice: Training examples can leave gaps**

REFERENCE · SECTION 13

Extra practice: Training examples can leave gaps

> Try the question before opening its worked explanation.

Read the supplied case and question

Training includes nine daylight leaf photos and one low-light photo.

What intended-use gap is present?

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

Low light is poorly represented. Add varied, correctly labelled low-light examples if that use is intended. Copies of one photo do not add equivalent variety.

Evidence or stage	Value or result
Daylight photos	9
Low-light photos	1

▼ Earlier wording, alternatives and picture transcripts

Week 22 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Training examples can leave gaps

Training includes nine daylight leaf photos and one low-light photo.

What intended-use gap is present?

Low light is poorly represented. Add varied, correctly labelled low-light examples if that use is intended. Copies of one photo do not add equivalent variety.

▼ Reference: original lesson checklist

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Name the target collection, selection list and cases actually assessed	Explain the deciding step, show a trace or test, and state one limit.
Use a coverage table to identify missing conditions before making a claim	Explain the deciding step, show a trace or test, and state one limit.
Propose a selection plan that addresses a stated gap without promising certainty	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: A fictional badge reader will be used in both bright and dim rooms.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional badge reader will be used in both bright and dim rooms.
- Which labelled step matches this detail? Select and record new cases from both dim groups before claiming **coverage**.
- 1. Target conditions
- 2. Current **sample**
- 3. Targeted follow-up
- 4. **Coverage gap**
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ Check the first explanation

After your attempt, compare with the supported explanation: Targeted follow-up Select and record new cases from both dim groups before claiming **coverage**.

Adding more bright cards can increase **sample** size without addressing the missing condition. A **coverage** check asks which relevant situations are absent, not merely how many rows exist.

Deliberately selecting dim cases helps investigate those conditions, but the resulting test should not be advertised as a random survey unless its selection really was random.

Changed-case question

- Changed-case practice: Add the two remaining bright plain cards and no dim cards. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. The **sample** becomes larger but still has no dim **coverage**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ Check the changed case

After your attempt, compare with the supported explanation: The **sample** becomes larger but still has no dim **coverage**. The missing lighting condition remains untested.

Repair and save your work

Review Good Data, Bad Data (ai-1/21). Goal: Check records against a stated format, unit and meaning before using them. Short practice: Which repairs are **evidence**-backed in the original case, and which row must remain unknown?.

Return to this week and explain what you changed.

Review Similarity (ai-1/19). Goal: Calculate similarity using explicitly selected features and a stated scoring rule. Short practice: Point to the matches supporting C's new score and explain why there is no tie..

Return to this week and explain what you changed.

Review Boundaries and 'Not Sure' (ai-1/16). Goal: Identify a specific source of uncertainty and explain how it affects a **prediction**. Short practice: Which original block needs a better width before accepting or rejecting it?

What should the reviewer do?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

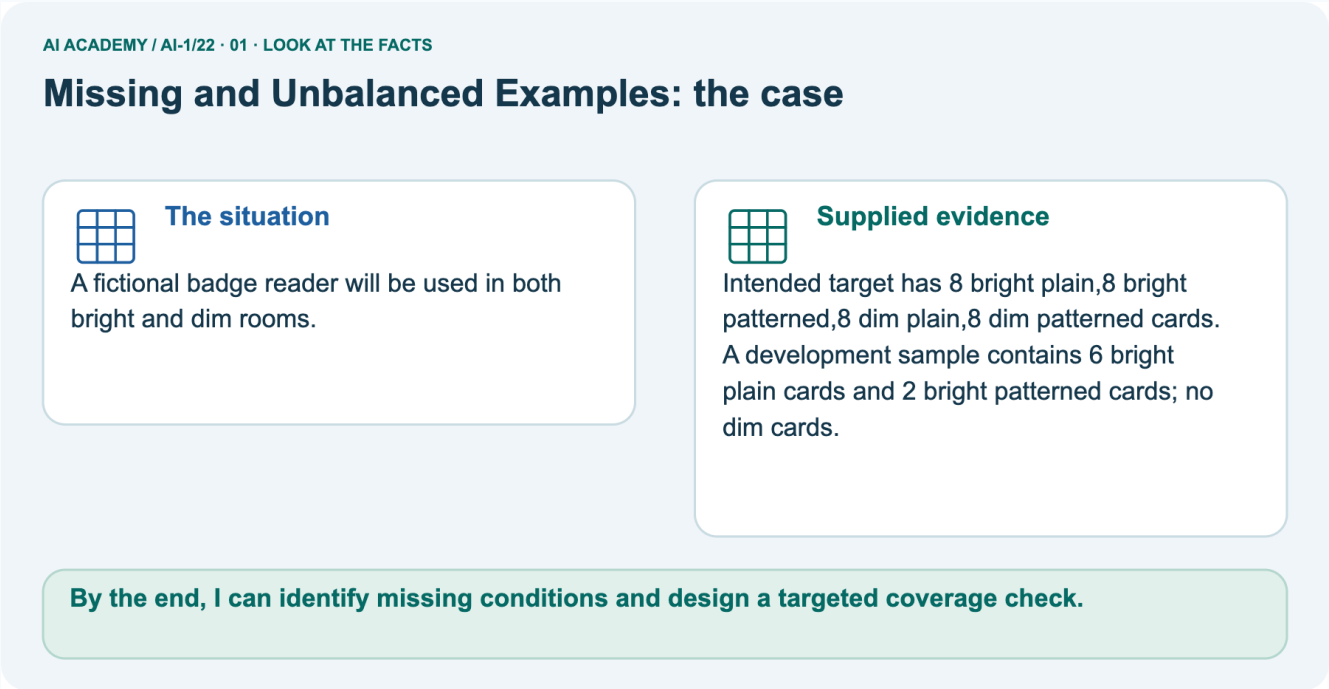
Keep your completed work

▼ Reference: the six-picture journey

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Missing and Unbalanced Examples: the case



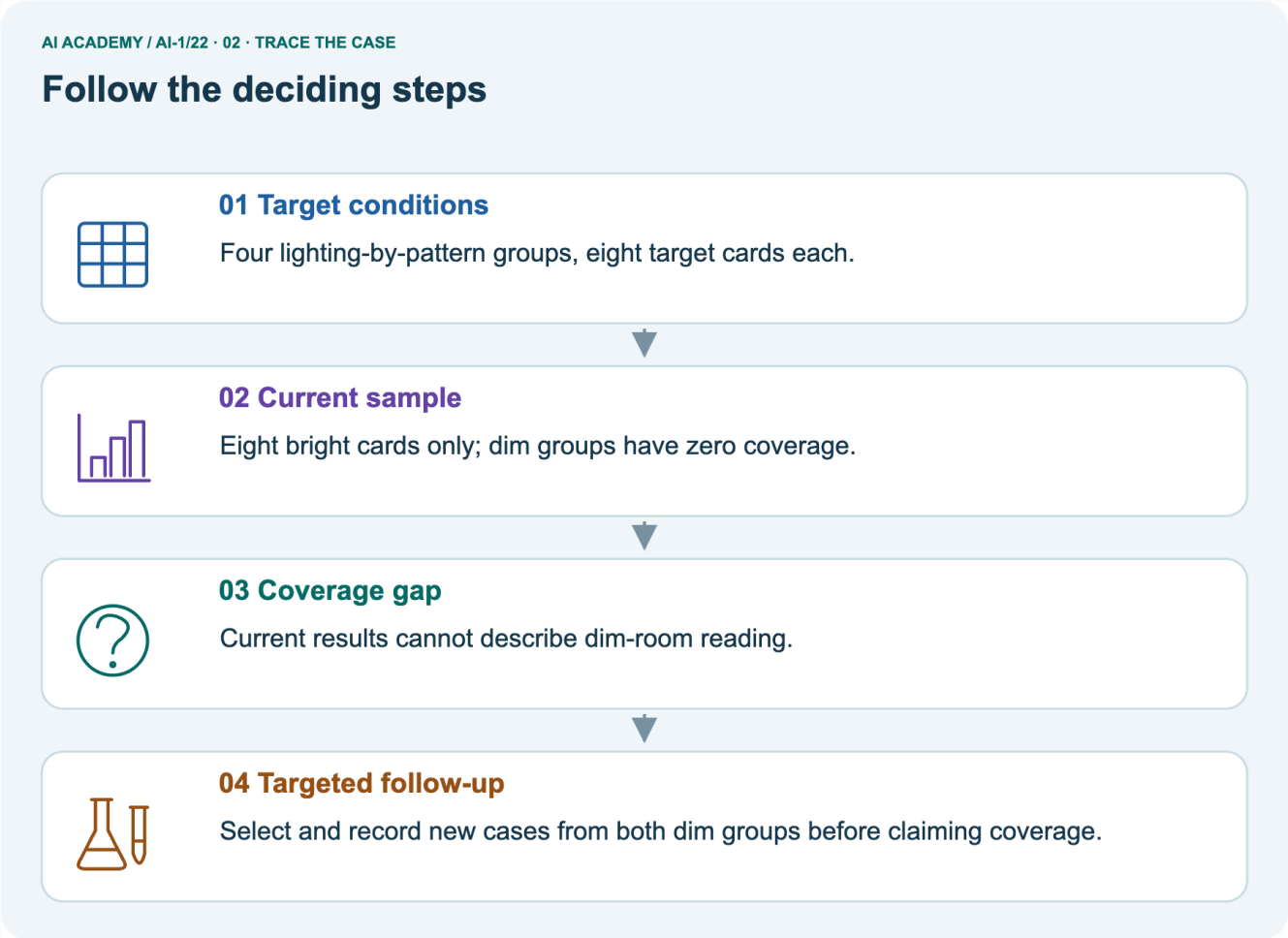
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional badge reader will be used in both bright and dim rooms. Supplied facts: Intended target has 8 bright plain,8 bright patterned,8 dim plain,8 dim patterned cards.

A development **sample** contains 6 bright plain cards and 2 bright patterned cards; no dim cards.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Target conditions: Four lighting-by-pattern groups, eight target cards each. → Current **sample**: Eight bright cards only; dim groups have zero **coverage**. → **Coverage gap**: Current results cannot describe dim-room reading. → Targeted follow-up: Select and record new cases from both dim groups before claiming **coverage**.

Point to each numbered stage and explain which recorded **input** it uses.

03. Target groups and development coverage



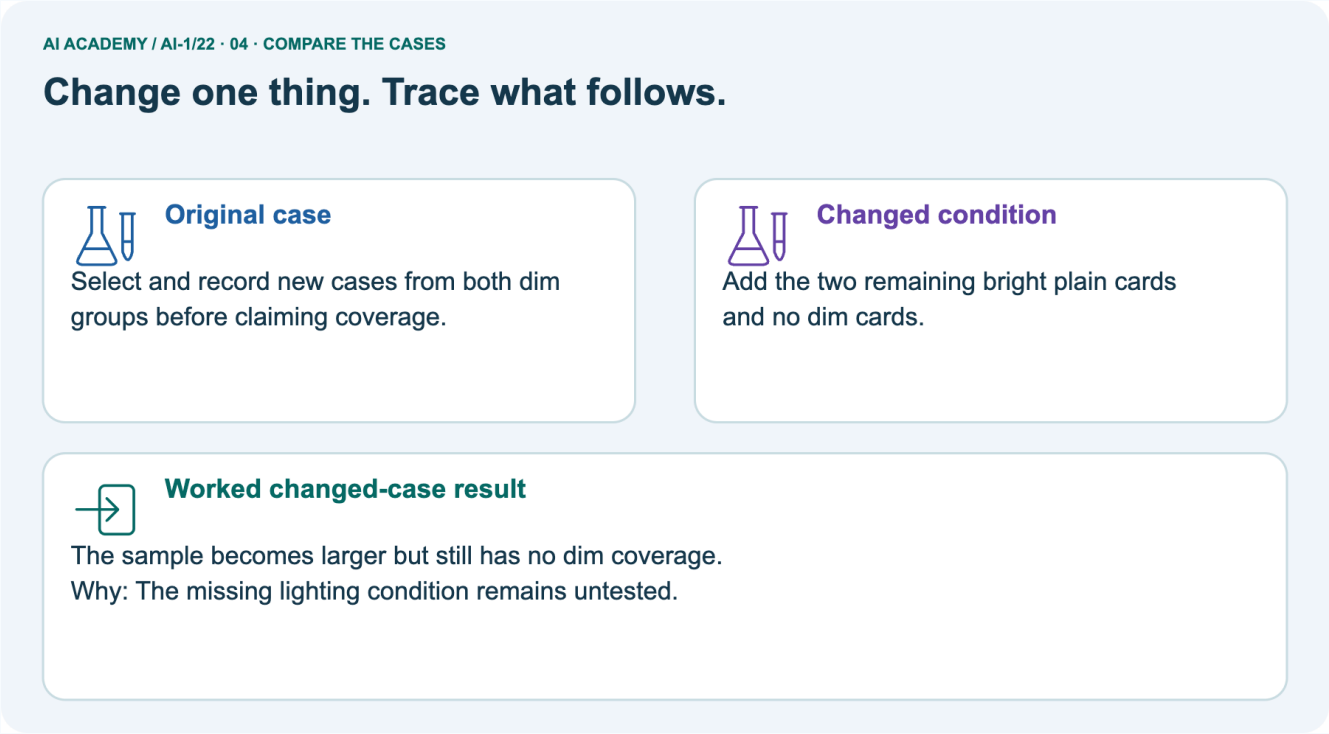
Follow the labelled picture with the matching explanation. Open the vector diagram

Target groups and development **coverage**. Bright plain: Target cards = 8, **Sample** cards = 6. Bright patterned: Target cards = 8, **Sample** cards = 2. Dim plain: Target cards = 8, **Sample** cards = 0.

Dim patterned: Target cards = 8, **Sample** cards = 0.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



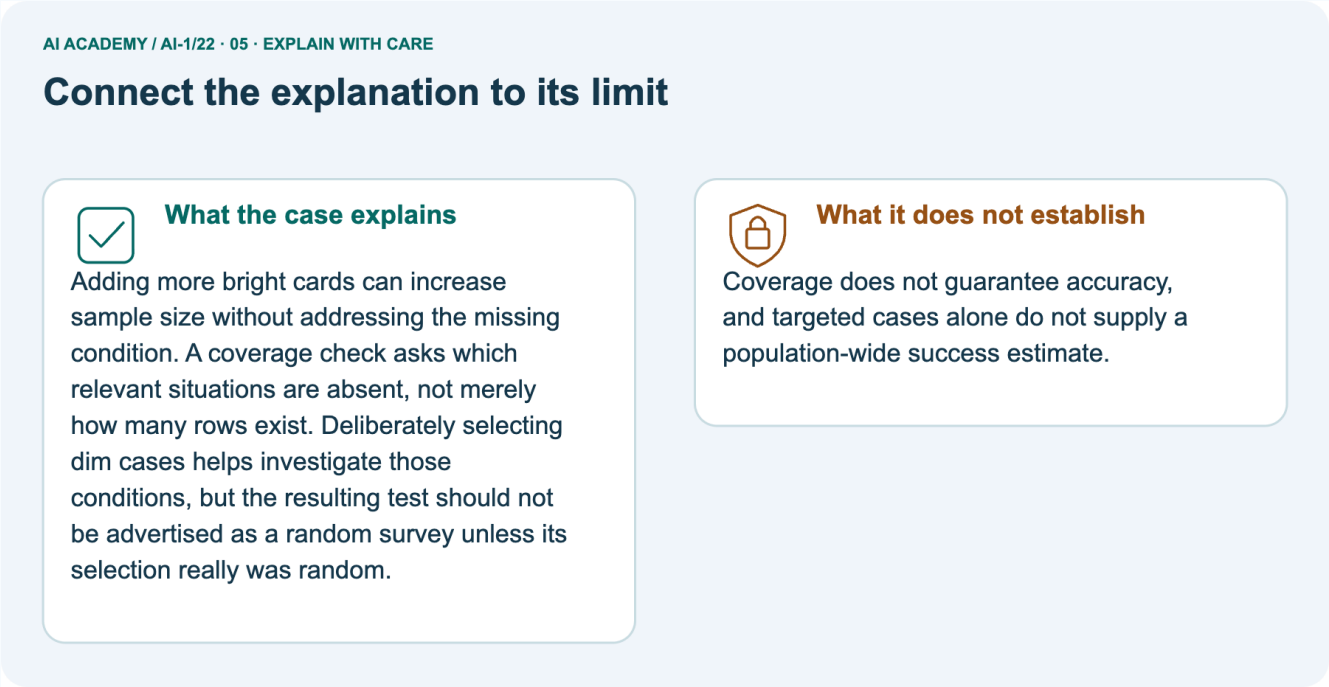
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Select and record new cases from both dim groups before claiming **coverage**.
Changed condition: Add the two remaining bright plain cards and no dim cards.

Expected result: The **sample** becomes larger but still has no dim **coverage**. Reason: The missing lighting condition remains untested.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

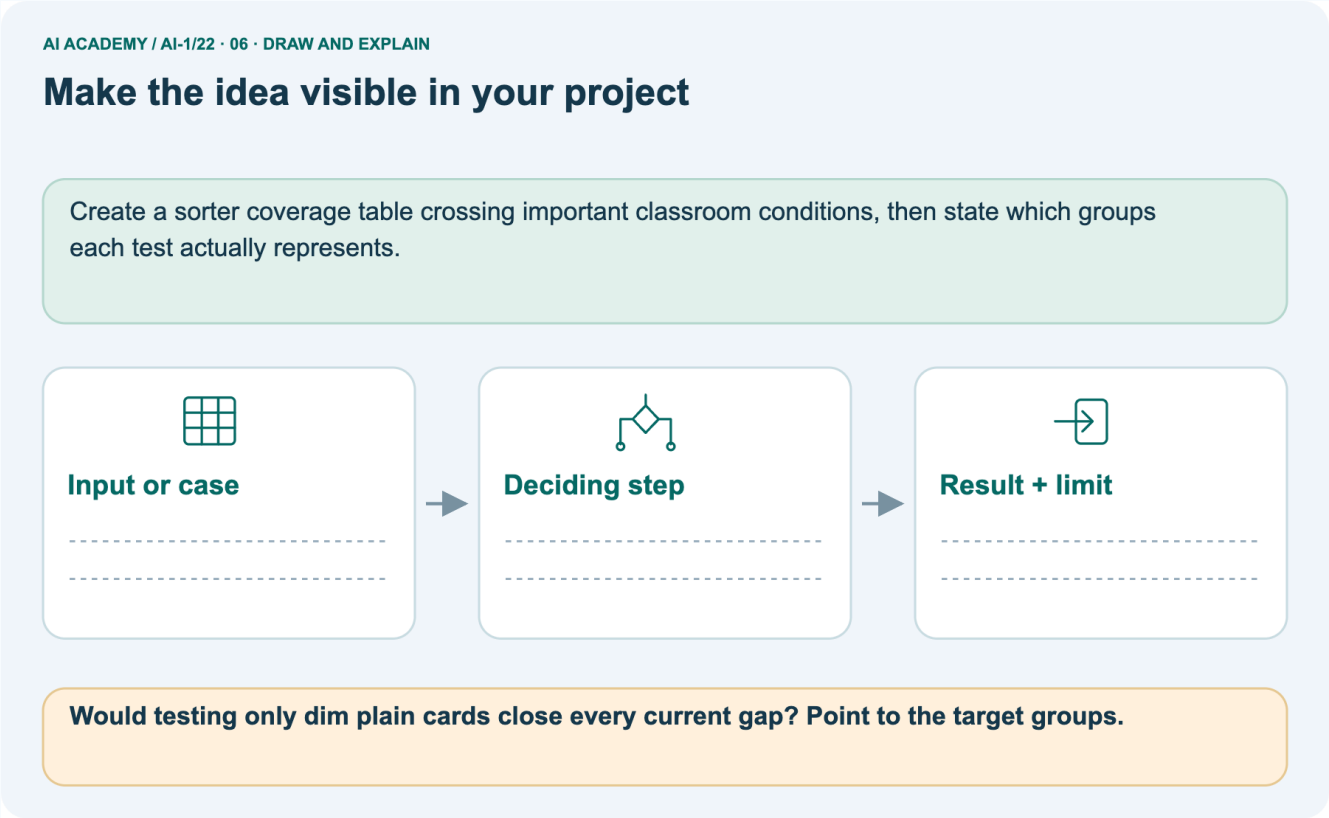
Adding more bright cards can increase **sample** size without addressing the missing condition. A **coverage** check asks which relevant situations are absent, not merely how many rows exist.

Deliberately selecting dim cases helps investigate those conditions, but the resulting test should not be advertised as a random survey unless its selection really was random.

Limit: **Coverage** does not guarantee accuracy, and targeted cases alone do not supply a population-wide success estimate.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Create a sorter **coverage** table crossing important classroom conditions, then state which groups each test actually represents. Draw your **input**, deciding step, result and limit.

Supported question: Would testing only dim plain cards close every current gap? Point to the target groups.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Missing and Unbalanced Examples: the case

02. Follow the deciding steps

03. Target groups and development coverage

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.